

$$\begin{cases} x_2 + x_1 y_1 + y_2^2 = 0 \\ x_1 - x_2 y_2 + y_1^2 = 0 \end{cases}$$

Assume that this system
defines (y_1, y_2) as
function of (x_1, x_2)
from an open ball
around $(x_1^0, x_2^0) = (-1, 1)$
to an open ball around
 $(y_1^0, y_2^0) = (1, 0)$
+ calculate derivative

1) Convert to standard form
for implicit function theorem

$$F(x_1, x_2, y_1, y_2) = \begin{bmatrix} x_2 + x_1 y_1 + y_2^2 \\ x_1 - x_2 y_2 + y_1^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Calculate Jacobian of F

$$D_{(y_1, y_2)} F \Big|_{\substack{(x_1^0, x_2^0) \\ (y_1^0, y_2^0)}} = \begin{bmatrix} \frac{\partial F_1}{\partial y_1} & \frac{\partial F_1}{\partial y_2} \\ \frac{\partial F_2}{\partial y_1} & \frac{\partial F_2}{\partial y_2} \end{bmatrix} = \begin{bmatrix} x_1 & 2y_1 \\ 2y_2 & -x_2 \end{bmatrix} \Big|_{\substack{(x_1^0, x_2^0) \\ (y_1^0, y_2^0)}}$$

$$= \begin{bmatrix} -1 & 2(0) \\ 2(0) & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Det $D_{(y_1, y_2)} F = 1$ so invertible
by implicit function theorem,

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_1(x_1, x_2) \\ y_2(x_1, x_2) \end{bmatrix} \leftarrow \begin{array}{l} \text{how does} \\ \text{this} \\ \text{follow} \\ \text{from Jacobian?} \end{array}$$

to calculate derivative

$$F(x_1, x_2, y_1(x_1, x_2), y_2(x_1, x_2)) = 0$$

$$\frac{\partial F_1}{\partial x_1} + \frac{\partial F_1}{\partial y_1} \frac{dy_1}{dx_1} + \frac{\partial F_1}{\partial y_2} \frac{dy_2}{dx_1} = 0$$

$$y_1 + x_1 \frac{dy_1}{dx_1} + 2y_2 \frac{dy_2}{dx_1} = 0$$

plug in
 (y_1, y_2)
 (x_1, x_2)

$$1 + (-1) \frac{dy_1}{dx_1} + 2(0) = 0 \Rightarrow \frac{dy_1}{dx_1} = 1$$

$$\frac{\partial F_2}{\partial x_1} + \frac{\partial F_2}{\partial y_1} \frac{\partial y_1}{\partial x_1} + \frac{\partial F_2}{\partial y_2} \frac{\partial y_2}{\partial x_1} = 0$$

$$1 + 2y \frac{\partial y_1}{\partial x_1} + (-x_2) \frac{\partial y_2}{\partial x_1} = 0$$

$$1 + 2 \frac{\partial y_2}{\partial x_1} - \frac{\partial y_2}{\partial x_1} = 0 \Rightarrow \frac{\partial y_2}{\partial x_1} = -1$$

repeat above
steps & clarify

$$D_x y = -[D_y F]^{-1} [D_x F]$$

Problem 6. Continuous bijections from compact sets

Let $K \subseteq \mathbb{R}^n$ be compact, and suppose

$$f: K \rightarrow Y \subseteq \mathbb{R}^m$$

is a continuous bijection.

Prove that

$$f^{-1}: Y \rightarrow K$$

is continuous.

Guided contradiction argument. Take a sequence

$$y_k \rightarrow y$$

and define

$$x_k = f^{-1}(y_k), \quad x = f^{-1}(y).$$

Suppose that $x_k \not\rightarrow x$. Show that there exist $\varepsilon > 0$ and a subsequence (x_{k_j}) such that

$$\|x_{k_j} - x\| \geq \varepsilon$$

for every j .

Use compactness to extract a further convergent subsequence

$$x_{k_{j_\ell}} \rightarrow x^*.$$

Now compare

$$f(x_{k_{j_\ell}}) \rightarrow f(x^*)$$

with

$$f(x_{k_{j_\ell}}) = y_{k_{j_\ell}} \rightarrow y.$$

Identify where uniqueness of limits is used and where injectivity of f is used.

$$K \subseteq \mathbb{R}^n \text{ compact}$$

$$f: K \rightarrow Y \subseteq \mathbb{R}^n \text{ bijective} \\ \text{+ continuous}$$

Prove $f^{-1}: Y \rightarrow K$ is continuous

wmf to show

$$y, \exists u \in Y: x_n = f^{-1}(y_n) \rightarrow f^{-1}(y) = u$$

want to show -

$$y_k \ni y \in Y : x_k = f^{-1}(y_k) \rightarrow f^{-1}(y) = x$$

suppose not

suppose $x_k \not\rightarrow x$:

$$\exists \epsilon_0 > 0 \quad \forall N \in \mathbb{N} : \exists k > N$$

$$\|x_k - x\| \geq \epsilon_0 \quad (*)$$

take $N=1 \Rightarrow x_{k_1}$ that satisfies $(*)$

$N=2 \Rightarrow x_{k_2}$ satisfies $(*)$

$\vdots$

$N=j \Rightarrow x_{k_j}$ satisfies $(*)$

then there is sequence

$$\{x_{k_j}\} : \|x_{k_j} - x\| \geq \epsilon_0 \quad \forall j$$

Then we have subsequence of

$$\{x_{k_j}\} \text{ called } \{x_{k_{j_0}}\} :$$

by
compactness
of K

$$x_{k_{j_0}} \rightarrow x^* \in X$$

$$\text{Then } f(x_{k_{j_0}}) \rightarrow f(x^*)$$

$$\text{but } f(x_{k_{j_0}}) = y_{k_{j_0}} \rightarrow y$$

$$\text{but } y = f(x)$$

then $f(x) = f(x^*)$ by
uniqueness of limit

$$f(x) = f(x^*) \Rightarrow x = x^*$$

uniqueness of limit
 bijective so $x = x^*$

random point

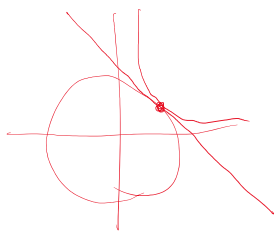
$$f^{-1}(A^c) = [f^{-1}(A)]^c$$

Newtonton:

Kuhn Tucker optimization

$$\max_{x \in D} f(x) \quad \text{s.t.} \quad g(x) \leq 0$$

using linear approximation in derivatives
 need solution in feasible region
 w/ tangency to objective funct



$$g(x) dx \leq 0$$

insufficient
 rule

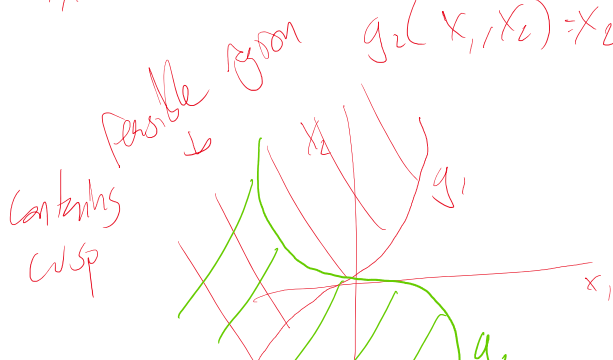
for constraint
 region

objective function needs to be differentiable

ex.

$$\max_{x_1, x_2} x_1 \quad \text{s.t.} \quad g_1(x_1, x_2) = x_1^3 - x_2 \leq 0$$

$$g_2(x_1, x_2) = x_2 + x_1^3 \leq 0$$





$$L = x_1 + \lambda_1(x_1^3 - x_2) + \lambda_2(x_1^3 + x_2)$$

Cases: $\lambda_1 = 0, \lambda_2 = 0$

$\lambda_1 > 0, \lambda_2 = 0$

$\lambda_1 = 0, \lambda_2 > 0$

$\lambda_1 > 0, \lambda_2 > 0$

Take $\lambda_1 = 0, \lambda_2 = 0$:

$L = x_1$, only FOC

$L' = 1 \neq 0$
no solution

Take $\lambda_1 > 0, \lambda_2 > 0$:

$L'_{x_1} = 1 + 3\lambda_1 x_1^2 + 3\lambda_2 x_1^2 = 0$ $L \neq 0$

but no solution

$L'_{x_2} = 0 - \lambda_1 + \lambda_2 = 0$

w/ this



$L'_{\lambda_1} = x_1^3 - x_2 = 0 \quad x_1^3 = x_2$ $x_1 = 0$

$\Rightarrow x_2 = 0$

$L'_{\lambda_2} = x_1^3 + x_2 = 0 \quad x_1^3 = -x_2$

Take $\lambda_1 > 0, \lambda_2 = 0$

$L_x = 1 + 3\lambda_1 x_1^2 + 3(0)x_1^2 = 0$

$$L_{x_1} = 1 + 3\lambda_1 x_1^2 + 3(0)x_1^2 = 0$$

$$L_{x_2} = -\lambda_1 + \lambda_2 = 0 \Rightarrow \lambda_1 = 0$$

contradiction
no solution

Take $\lambda_1 = 0, \lambda_2 > 0$

$$L_{x_2} = \lambda_2 = 0 \text{ contradiction}$$

no solution

So Kuhn Tucker give no solution
because gradient of feasible
region not defined
need to check constraint qualification
gradient of constraints

$$\nabla g_1(x) = \begin{bmatrix} 3x_1^2 \\ -1 \end{bmatrix}$$

$$\nabla g_2(x) = \begin{bmatrix} 3x_1^2 \\ 1 \end{bmatrix}$$

if $x_1 = 0$, ∇g_1 & ∇g_2
linearly dependent

so if $x_1 = 0$, constraints
collapse

$$\text{w/ } x_1 = 0, \quad \begin{matrix} -x_2 \leq 0 \\ x_2 \leq 0 \end{matrix} \Rightarrow x_2 = 0$$

thus $x_1=0, x_2=0$ is optimized

log linearization

2 versions: equal to zero
& not equal to zero

part 1:

$$f: \mathbb{R}_{++}^n \rightarrow \mathbb{R}^2 \quad \text{is } C^2$$

$$f(x_1, \dots, x_n) = 0 \quad \forall x \in D \quad \text{usually}$$

equilibrium

$\bar{x}$ steady state equilibrium

we get $f_1, \dots, f_n$
but process
is similar

$$f(\bar{x}_1, \dots, \bar{x}_n) = 0$$

change of variables

$$\text{define } \hat{x}_i = \log x_i - \log \bar{x}_i$$

percentage
deviation
from $\bar{x}$

$$\text{then } x_i = \bar{x}_i e^{\hat{x}_i}$$

$$= \bar{x}_i \exp\{\log x_i - \log \bar{x}_i\}$$

$$= \bar{x}_i \frac{x_i}{\bar{x}_i} = x_i$$

define:

$$g(\hat{x}_1, \dots, \hat{x}_n) = f(\bar{x}_1 e^{\hat{x}_1}, \dots, \bar{x}_n e^{\hat{x}_n})$$

$$\text{w/ } g(0) = f(\bar{x}) = 0$$

use Taylor's approximation for $\hat{x}$ near 0

$$g(\hat{x}) \approx g(0) + \sum_{i=1}^n \frac{\partial g}{\partial \hat{x}_i}(0) \hat{x}_i + r(\hat{x})$$

vanishing
remainder

$$g(\hat{x}) = 0$$

& since $g(0) = 0$,

we're at equilibrium

$\hat{x}_1, \hat{x}_2, \dots, \hat{x}_n$

the y and u
steady state conditions
for market clearing

$$\sum_{i=1}^n \frac{\partial g}{\partial x_i}(0) \hat{x}_i = 0$$

evaluate derivatives
using chain rule

$$\frac{\partial g}{\partial \hat{x}_i}(\hat{x}) = f_{x_i}(\bar{x}^{\hat{x}_1}, \dots, \bar{x}^{\hat{x}_n}) \cdot \frac{\partial(\bar{x}_i e^{\hat{x}_i})}{\partial \hat{x}_i}$$

$$= f_{x_i}(\bar{x}^{\hat{x}_1}, \dots, \bar{x}^{\hat{x}_n}) \cdot \bar{x}_i e^{\hat{x}_i}$$

$$= f_{x_i}(\bar{x}_1 e^{\hat{x}_1}, \dots, \bar{x}_n e^{\hat{x}_n}) \bar{x}_i$$

thus log linearized equation

$$\sum_{i=1}^n \bar{x}_i f_{x_i}(\bar{x}) \hat{x}_i = 0$$

Part 2: alternative approach
to log linearization

$$f(x^*) \neq 0 \quad \text{w/ } x^* \in \mathbb{R}^n$$

$$\text{define elasticity } \eta_i = \frac{f_i(x^*)}{f(x^*)} x_i^*$$

which we
just change $\frac{1}{f} \cdot \frac{x_i}{f} = \frac{1}{f} \cdot \frac{\Delta f}{\Delta x_i}$
in f resulting Δx_i f $1/\Delta x_i$
from f_i
divided by
change in x

$$f(x) = f(x^*) \left[1 + \eta_1 \hat{x}_1 + \dots + \eta_n \hat{x}_n \right]$$

Define log-derivation of function
at $x^* = (x_1^*, \dots, x_n^*)$

Let the log-likelihood be

$$f \text{ and } x^* = (x_1^*, \dots, x_n^*)$$

$$\text{s.t. } f(x^*) \neq 0$$

$$\text{h.s. } \hat{f}(x) = \ln(f(x)/f(x^*))$$

$$\Rightarrow \frac{f(x) - f(x^*)}{f(x)}$$

$$\approx \ln(f(x)/f(x^*))$$

$$\text{s.t. } \hat{f}(x) \approx \eta_1 \hat{x}_1 + \dots + \eta_n \hat{x}_n$$

Shortcuts

$$1. dx$$

$$\hat{dx} = \eta_1 \hat{x} \quad \text{w/ } \eta = \frac{dx^*}{f(x^*)} = \frac{dx^*}{dx^*} = 1$$

$$2. x_1 + x_2$$

$$\hat{x}_1 + \hat{x}_2 = \eta_1 \hat{x}_1 + \eta_2 \hat{x}_2 \quad \text{w/ } \eta_1 = \frac{1 \cdot x_1^*}{x_1^* + x_2^*}$$

$$\eta_2 = \frac{1 \cdot x_2^*}{x_1^* + x_2^*}$$

$$3. x_1 x_2$$

$$\hat{x}_1 \hat{x}_2 = \hat{x}_1 + \hat{x}_2 \quad \text{w/ } \eta_1 = \frac{x_2^*}{x_1^*} = 1$$

$$\eta_2 = \dots = 1$$

$$4. x^d$$

TBD

$$5. t(x)$$

$$G, C \in \mathbb{R}$$

Example:

$$Y = A K^{\alpha} L^{1-\alpha}$$

$$\hat{Y} = ? \quad \eta_A = \frac{K^{\alpha} L^{1-\alpha} A}{Y} = \frac{K^{\alpha} L^{1-\alpha} A}{A K^{\alpha} L^{1-\alpha}} = 1$$

$$\eta_K = \frac{A \cdot \alpha K^{\alpha-1} L^{1-\alpha}}{A K^{\alpha} L^{1-\alpha}} K = \alpha$$

$$\eta_L = \frac{A K^{\alpha} (1-\alpha) L^{-\alpha}}{A K^{\alpha} L^{1-\alpha}} L = (1-\alpha)$$

$$\hat{Y} = \hat{A} + \alpha \hat{K} + (1-\alpha) \hat{L}$$

$$\bar{Y} e^{\hat{Y}} = \bar{A} e^{\hat{A}} (\bar{K} e^{\hat{K}})^{\alpha} (\bar{L} e^{\hat{L}})^{1-\alpha}$$

$$\bar{Y} e^{\hat{Y}} = \bar{Y} e^{\hat{A} + \alpha \hat{K} + (1-\alpha) \hat{L}}$$

$$\hat{Y} = \hat{A} + \alpha \hat{K} + (1-\alpha) \hat{L}$$

Can't we
just take log ?

just take log
+ derivative?